STAT5147 - Homework 5

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1 Problem 1 Answers

We are given a population of $N = 9$ executives, consisting of 4 married, 3 never-married, and 2 divorced executives. A random sample of $n = 3$ executives is selected without replacement.

Let Y_1 denote the number of married executives and Y_2 denote the number of never-married executives in the sample.

Since the selection is made without replacement, (Y_1, Y_2) follows a multivariate hypergeometric distribution.

For a hypergeometric random variable Y with population size N, number of successes K, and sample size n, the expectation and variance are

$$\begin{aligned} E(Y) &= n\frac{K}{N}, \\ \text{Var}(Y) &= n\frac{K}{N}\left(1 - \frac{K}{N}\right)\frac{N-n}{N-1}. \end{aligned}$$

1.1 (a) Expectation and Variance of Y_1

There are $K = 4$ married executives among the $N = 9$ executives, and $n = 3$ are selected.

The expected number of married executives is

$$\begin{aligned} E(Y_1) &= n\frac{K}{N} \\ &= 3\left(\frac{4}{9}\right) \\ &= \frac{4}{3}. \end{aligned}$$

The variance is

$$\begin{aligned} \text{Var}(Y_1) &= n\frac{K}{N}\left(1 - \frac{K}{N}\right)\frac{N-n}{N-1} \\ &= 3\left(\frac{4}{9}\right)\left(\frac{5}{9}\right)\left(\frac{6}{8}\right) \\ &= \frac{5}{9}. \end{aligned}$$

1.2 (b) Expectation and Variance of Y_2

There are $K = 3$ never-married executives among the $N = 9$ executives, and $n = 3$ are selected.

The expected number of never-married executives is

$$\begin{aligned} E(Y_2) &= n\frac{K}{N} \\ &= 3\left(\frac{3}{9}\right) \\ &= 1. \end{aligned}$$

The variance is

$$\begin{aligned} \text{Var}(Y_2) &= n\frac{K}{N}\left(1 - \frac{K}{N}\right)\frac{N-n}{N-1} \\ &= 3\left(\frac{3}{9}\right)\left(\frac{6}{9}\right)\left(\frac{6}{8}\right) \\ &= \frac{1}{2}. \end{aligned}$$

1.3 (c) Covariance Between Y_1 and Y_2

By definition,

$$\text{Cov}(Y_1, Y_2) = E(Y_1Y_2) - E(Y_1)E(Y_2).$$

For two distinct categories in a multivariate hypergeometric distribution,

$$E(Y_1Y_2) = \frac{n(n-1)K_1K_2}{N(N-1)},$$

where $K_1 = 4$ is the number of married executives and $K_2 = 3$ is the number of never-married executives.

Therefore,

$$\begin{aligned} E(Y_1Y_2) &= \frac{3(3-1)(4)(3)}{9(9-1)} \\ &= \frac{72}{72} \\ &= 1. \end{aligned}$$

Using the expectations from parts (a) and (b),

$$\begin{aligned} \text{Cov}(Y_1, Y_2) &= E(Y_1Y_2) - E(Y_1)E(Y_2) \\ &= 1 - \left(\frac{4}{3}\right)(1) \\ &= -\frac{1}{3}. \end{aligned}$$

The covariance is negative because selecting more married executives leaves fewer positions available for never-married executives in the sample.

1.4 (d) Expectation and Variance of $Y_1 + Y_2$

Let $S = Y_1 + Y_2$ denote the total number of married and never-married executives selected.

By linearity of expectation,

$$\begin{aligned} E(S) &= E(Y_1 + Y_2) \\ &= E(Y_1) + E(Y_2) \\ &= \frac{4}{3} + 1 \\ &= \frac{7}{3}. \end{aligned}$$

For the variance of a sum of two random variables,

$$\mathrm{Var}(Y_1 + Y_2) = \mathrm{Var}(Y_1) + \mathrm{Var}(Y_2) + 2\,\mathrm{Cov}(Y_1, Y_2).$$

Substituting the results from parts (a), (b), and (c),

$$\begin{aligned} \mathrm{Var}(S) &= \frac{5}{9} + \frac{1}{2} + 2\left(-\frac{1}{3}\right) \\ &= \frac{10}{18} + \frac{9}{18} - \frac{12}{18} \\ &= \frac{7}{18}. \end{aligned}$$

As a verification, there are $4 + 3 = 7$ executives who are either married or never-married. Thus, $S \sim \mathrm{Hypergeometric}(9, 7, 3)$.

Applying the hypergeometric formulas directly,

$$E(S) = 3\left(\frac{7}{9}\right) = \frac{7}{3},$$

$$\mathrm{Var}(S) = 3\left(\frac{7}{9}\right)\left(\frac{2}{9}\right)\left(\frac{6}{8}\right) = \frac{7}{18}.$$

Both results agree with the calculations above.

2 Problem 2 Answers

We are given the joint probability density function

$$f(y_1, y_2) = \begin{cases} 4y_1 y_2, & 0 \le y_1 \le 1,\ 0 \le y_2 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

First, we determine the marginal probability density functions of Y_1 and Y_2.

For Y_1,

$$\begin{aligned} f_{Y_1}(y_1) &= \int_0^1 f(y_1, y_2)\, dy_2 \\ &= \int_0^1 4y_1 y_2\, dy_2 \\ &= 4y_1 \left[\frac{y_2^2}{2}\right]_0^1 \\ &= 2y_1, \qquad 0 \le y_1 \le 1. \end{aligned}$$

Similarly, for Y_2,

$$\begin{aligned} f_{Y_2}(y_2) &= \int_0^1 f(y_1, y_2)\, dy_1 \\ &= \int_0^1 4y_1 y_2\, dy_1 \\ &= 4y_2 \left[\frac{y_1^2}{2}\right]_0^1 \\ &= 2y_2, \qquad 0 \le y_2 \le 1. \end{aligned}$$

2.1 (a) Expected Values of Y_1 and Y_2

By definition, the expected value of a continuous random variable Y is

$$E(Y) = \int_{-\infty}^{\infty} y f_Y(y)\, dy.$$

Therefore,

$$\begin{aligned} E(Y_1) &= \int_0^1 y_1 f_{Y_1}(y_1)\, dy_1 \\ &= \int_0^1 y_1 (2y_1)\, dy_1 \\ &= 2\int_0^1 y_1^2\, dy_1 \\ &= 2\left[\frac{y_1^3}{3}\right]_0^1 \\ &= \frac{2}{3}. \end{aligned}$$

Similarly,

$$
\begin{aligned}
E(Y_2) &= \int_0^1 y_2 f_{Y_2}(y_2)\, dy_2 \\
&= \int_0^1 y_2 (2y_2)\, dy_2 \\
&= 2\int_0^1 y_2^2\, dy_2 \\
&= 2\left[\frac{y_2^3}{3}\right]_0^1 \\
&= \frac{2}{3}.
\end{aligned}
$$

Hence,

$$
E(Y_1) = \frac{2}{3}, \qquad E(Y_2) = \frac{2}{3}.
$$

2.2 (b) Variances of Y_1 and Y_2

The variance of a random variable Y is

$$
V(Y) = E(Y^2) - [E(Y)]^2.
$$

First, we calculate $E(Y_1^2)$:

$$
\begin{aligned}
E(Y_1^2) &= \int_0^1 y_1^2 f_{Y_1}(y_1)\, dy_1 \\
&= \int_0^1 y_1^2 (2y_1)\, dy_1 \\
&= 2\int_0^1 y_1^3\, dy_1 \\
&= 2\left[\frac{y_1^4}{4}\right]_0^1 \\
&= \frac{1}{2}.
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
V(Y_1) &= E(Y_1^2) - [E(Y_1)]^2 \\
&= \frac{1}{2} - \left(\frac{2}{3}\right)^2 \\
&= \frac{1}{2} - \frac{4}{9} \\
&= \frac{1}{18}.
\end{aligned}
$$

Similarly, for Y_2,

$$\begin{aligned} E(Y_2^2) &= \int_0^1 y_2^2 f_{Y_2}(y_2)\, dy_2 \\ &= 2\int_0^1 y_2^3\, dy_2 \\ &= 2\left[\frac{y_2^4}{4}\right]_0^1 \\ &= \frac{1}{2}. \end{aligned}$$

Thus,

$$\begin{aligned} V(Y_2) &= E(Y_2^2) - [E(Y_2)]^2 \\ &= \frac{1}{2} - \left(\frac{2}{3}\right)^2 \\ &= \frac{1}{18}. \end{aligned}$$

Hence,

$$V(Y_1) = \frac{1}{18}, \qquad V(Y_2) = \frac{1}{18}.$$

2.3 (c) Expected Value of $Y_1 - Y_2$

Using the linearity of expectation,

$$\begin{aligned} E(Y_1 - Y_2) &= E(Y_1) - E(Y_2) \\ &= \frac{2}{3} - \frac{2}{3} \\ &= 0. \end{aligned}$$

Therefore,

$$E(Y_1 - Y_2) = 0.$$

2.4 (d) Covariance Between Y_1 and Y_2

By definition,

$$\mathrm{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2).$$

First, we calculate $E(Y_1 Y_2)$ using the joint probability density function:

$$
\begin{aligned}
E(Y_1Y_2) &= \int_0^1 \int_0^1 y_1 y_2 f(y_1, y_2)\, dy_2\, dy_1 \\
&= \int_0^1 \int_0^1 4y_1^2 y_2^2\, dy_2\, dy_1 \\
&= 4\int_0^1 y_1^2 \left(\int_0^1 y_2^2\, dy_2 \right) dy_1 \\
&= 4\int_0^1 y_1^2 \left[\frac{y_2^3}{3} \right]_0^1 dy_1 \\
&= \frac{4}{3}\int_0^1 y_1^2\, dy_1 \\
&= \frac{4}{3}\left[\frac{y_1^3}{3} \right]_0^1 \\
&= \frac{4}{9}.
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
\mathrm{Cov}(Y_1, Y_2) &= E(Y_1Y_2) - E(Y_1)E(Y_2) \\
&= \frac{4}{9} - \left(\frac{2}{3}\right)\left(\frac{2}{3}\right) \\
&= \frac{4}{9} - \frac{4}{9} \\
&= 0.
\end{aligned}
$$

Hence,

$$
\mathrm{Cov}(Y_1, Y_2) = 0.
$$

A covariance of zero indicates that Y_1 and Y_2 are uncorrelated, meaning that there is no linear relationship between the two variables.

Although zero covariance does not necessarily imply independence in general, we can establish independence in this particular case.

Using the marginal density functions,

$$
\begin{aligned}
f_{Y_1}(y_1) f_{Y_2}(y_2) &= (2y_1)(2y_2) \\
&= 4y_1y_2 \\
&= f(y_1, y_2),
\end{aligned}
$$

for $0 \leq y_1, y_2 \leq 1$. Both sides are zero outside this region.

Since the joint density is the product of the marginal densities, Y_1 and Y_2 are independent.

2.5 (e) Variance of $Y_1 + Y_2$

The variance of the sum of two random variables is

$$V(Y_1 + Y_2) = V(Y_1) + V(Y_2) + 2\,\mathrm{Cov}(Y_1, Y_2).$$

From parts (b) and (d), we have

$$V(Y_1) = V(Y_2) = \frac{1}{18}$$

and

$$\mathrm{Cov}(Y_1, Y_2) = 0.$$

Therefore,

$$\begin{aligned} V(Y_1 + Y_2) &= V(Y_1) + V(Y_2) + 2\,\mathrm{Cov}(Y_1, Y_2) \\ &= \frac{1}{18} + \frac{1}{18} + 2(0) \\ &= \frac{2}{18} \\ &= \frac{1}{9}. \end{aligned}$$

Hence,

$$V(Y_1 + Y_2) = \frac{1}{9}.$$